SET 20: STOCK ANALYSIS

Dataset	Stock Prices
Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	AKILA S
Register Number	192424331
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Day	Saturday

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

PROGRAM 1: Line Chart of Stock Prices

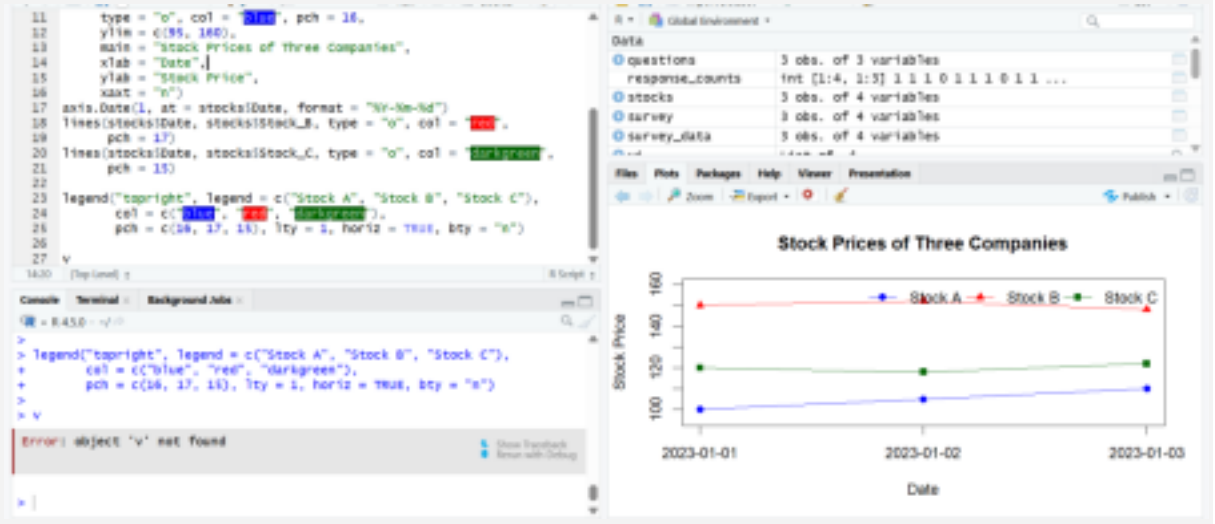
Question: Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

```
# Dataset
stocks <- data.frame(
 Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
Stock_A = c(100, 105, 110),
 Stock_B = c(150, 152, 148),
 Stock_C = c(120, 118, 122)
)

# Line Chart
plot(stocks$Date, stocks$Stock_A,
 type = "o", col = "blue", pch = 16,
 ylim = c(95, 160),
 main = "Stock Prices of Three Companies",
 xlab = "Date",
 ylab = "Stock Price",
 xaxt = "n")
axis.Date(1, at = stocks$Date, format = "%Y-%m-%d")
 lines(stocks$Date, stocks$Stock_B, type = "o", col = "red",
pch = 17)
```

```
lines(stocks$Date, stocks$Stock_C, type = "o", col = "darkgreen",
pch = 15)

legend("topright", legend = c("Stock A", "Stock B", "Stock C"),
col = c("blue", "red", "darkgreen"),
 pch = c(16, 17, 15), lty = 1, horiz = TRUE, bty = "n")
```



Output 1: Line Chart of Stock Prices

Observation: The line chart compares the three stocks over 1–3 January 2023. Stock A rises steadily from 100 to 110, Stock B stays the highest (148–152) with a small dip on the last day, and Stock C falls slightly on 2 January before recovering to 122.

PROGRAM 2: Bar Chart of Daily Percentage Change (Stock A)

Question: Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

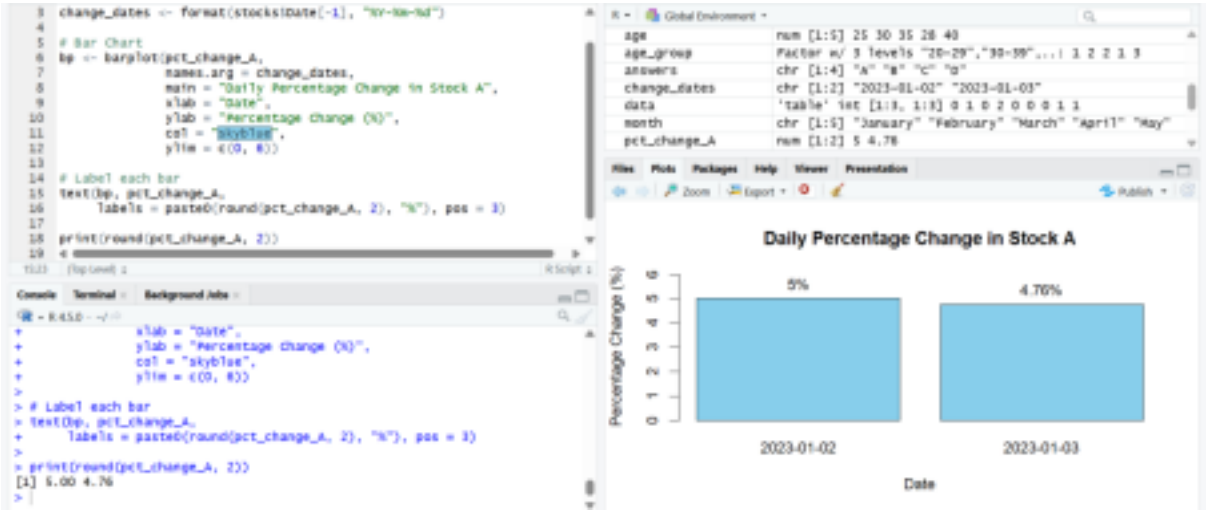
```
# Daily percentage change for Stock A
pct_change_A <- diff(stocks$Stock_A) / head(stocks$Stock_A, -1) * 100
change_dates <- format(stocks$Date[-1], "%Y-%m-%d")

# Bar Chart
bp <- barplot(pct_change_A,
 names.arg = change_dates,
 main = "Daily Percentage Change in Stock A",  xlab =
"Date",
 ylab = "Percentage Change (%)",
 col = "skyblue",
 ylim = c(0, 6))

# Label each bar
text(bp, pct_change_A,
 labels = paste0(round(pct_change_A, 2), "%"), pos = 3)

print(round(pct_change_A, 2))
```

```
> print(round(pct_change_A, 2))
[1] 5.00 4.76
```



Output 2: Daily Percentage Change in Stock A

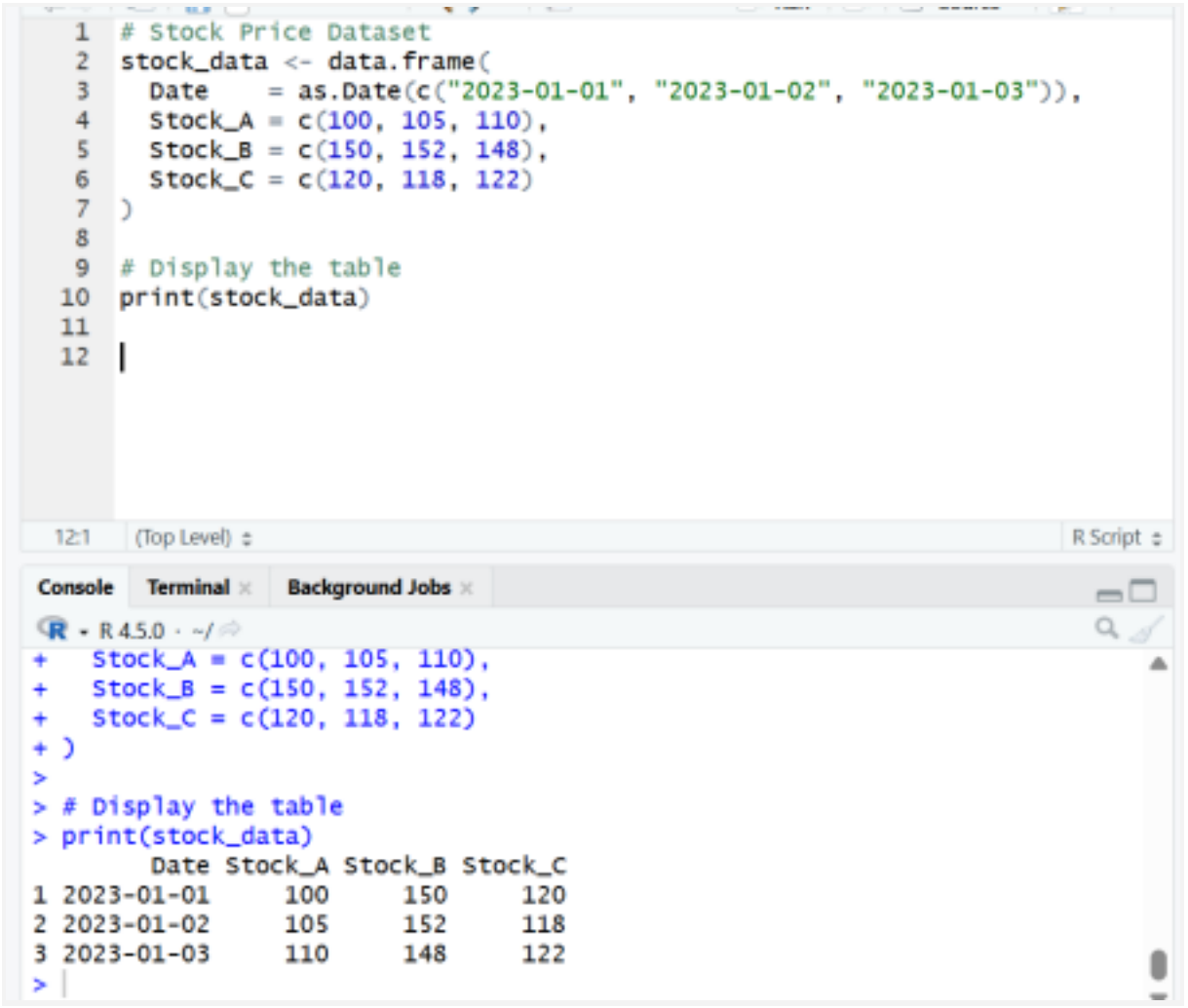
Observation: Stock A increased by 5.00% on 2 January 2023 (100 to 105) and by 4.76% on 3 January 2023 (105 to 110). No change is shown for 1 January because there is no earlier price to compare with.

PROGRAM 3: Stock Price Data Table

Question: Build a table to display the stock price data for each company over the given period. Label the table elements.

```
# Stock Price Dataset
stock_data <- data.frame(
 Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
Stock_A = c(100, 105, 110),
 Stock_B = c(150, 152, 148),
 Stock_C = c(120, 118, 122)
)

# Display the table
print(stock_data)
```



Output 3: Stock Price Data Table

Observation: The table displays the Date and the prices of Stock A, Stock B and Stock C for each of the three days.

PROGRAM 4: Interactive Tableau Dashboard

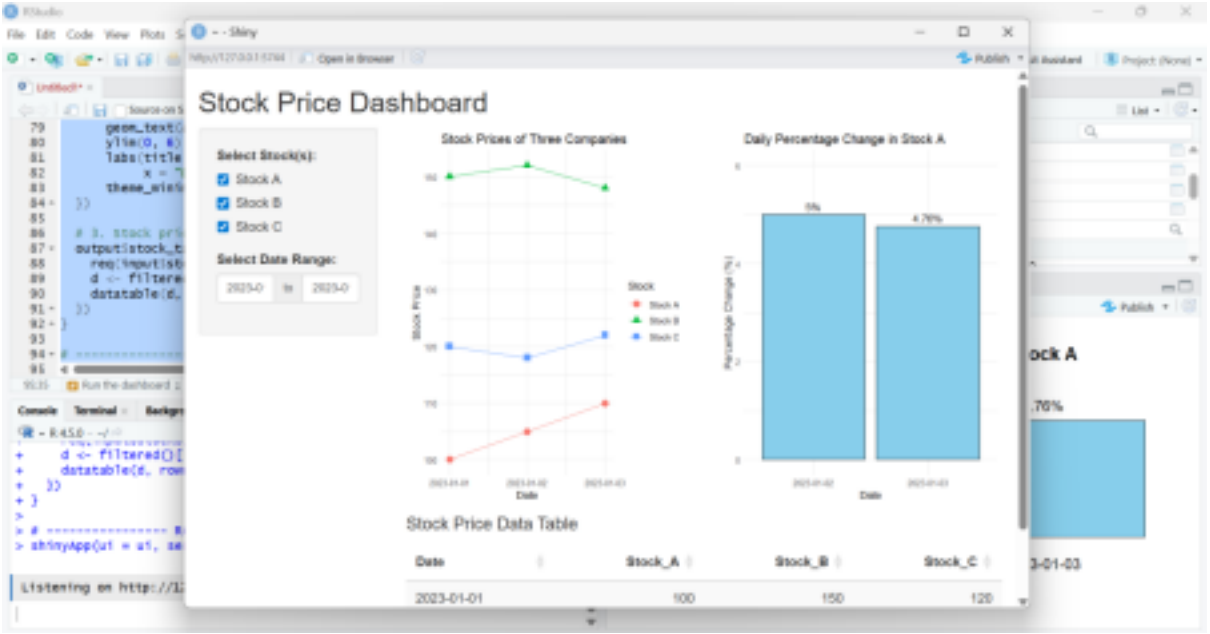
Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Tableau Steps

1. Open Tableau and connect to the Stock Prices dataset.
2. Verify the fields Date, Stock A, Stock B and Stock C.
3. Select the three stock columns, right-click and choose Pivot, then rename the new fields to Stock and Price.
4. Create a worksheet for the line chart: drag Date to Columns (continuous day), Price to Rows and Stock to Color; select Line in the Marks card.
5. Add the title “Stock Prices of Three Companies” and label the axes.
6. Create a second worksheet: drag Date to Columns and SUM(Price) to Rows, filter Stock to “Stock A”, right-click SUM(Price) and choose Quick Table Calculation > Percent

Difference; select Bar in the Marks card.

7. Add the title “Daily Percentage Change in Stock A” and label the axes.
8. Create a third worksheet: drag Date to Rows, Stock to Columns and Price to Text to form the stock price table.
9. Create a new Dashboard and add the line chart, bar chart and table.
10. Add the title “Stock Price Dashboard”.
11. Use Stock or a Date range as a filter (click “Use as Filter” or “Show Filter”) to make the dashboard interactive.
12. Test the dashboard by selecting a stock/date range and checking the linked views.



Output 4: Stock Price Dashboard

Observation: The Tableau dashboard combines the stock price line chart, the daily percentage change bar chart and the stock price table, allowing interactive exploration of the stock prices.

Procedure

1. Create the stock price data in R with the dates stored as Date values.
2. Create the line chart for Stock A, B and C using plot() and lines(), and add a legend. 3. Calculate the daily percentage change of Stock A using diff() and create the bar chart using barplot().
4. Create a data frame containing the Date and the three stock prices and display it as a table.
5. Open Tableau and import the Stock Prices dataset.
6. Pivot the stock columns and create the line chart and the percentage change bar chart in Tableau.
7. Create the stock price table in Tableau.
8. Combine the three visualizations into an interactive dashboard and add a suitable title. 9. Test the dashboard using the available filter/interactivity.

Result

The Stock Prices data was successfully visualized using a line chart, daily percentage change bar chart, stock price data table, and an interactive Tableau dashboard.